

QUESTION-TO-SLIDE MAPPING

SECTION A - LECTURE 08

Designing Microclimate

Course	BECM 3115 - Climate and Architectural Design
Lecture deck	8.0 - Designing Microclimate (8 PDF pages/slides)
Question bank	Regular term-final examinations, 2016-2024 (8 supplied papers; 2019 not present)
Method	Question content is mapped strictly to the eight-slide Lecture 08 deck. The deck is authoritative for climatic effects and design solutions for roof, floor, walls, windows, ventilators, doors, balcony, terrace, overhang and clerestory, plus the six design-tool groups. Solar-shading calculations and detailed thermal-property equations are not inferred from the slide images.

RESULT

Lecture 08 is the highest-value envelope lecture in Section A. The same building-envelope question appears in 2018, 2021, 2022, 2023 and 2024 with 10-22 marks. Slide 7 also directly answers the 2018 design-tools question. This is a compact 8-slide deck with unusually high marks-per-slide.

Prepared as part of the cumulative lecture-by-lecture question bank analysis.

1. Executive Summary

Lecture 08 treats the building envelope as the filter between climate and indoor microclimate. It explains how heat, humidity, air movement and solar/rain exposure affect roofs, floors, walls, windows and other envelope elements, then gives design responses such as reflectance, insulation, cavities, waterproofing, DPC, shading, ventilation and clerestories. The final content slide groups microclimate-control tools into materials, orientation, openings, architectural design, micro-planning and semi-mechanized techniques.

8
Slides reviewed

12
Question links

6
Exact matches

2
Direct matches

4
Partial matches

132
Mapped marks

Lecture structure and exam value

Slides	Lecture block	Historical signals	Exam value
2	Roof: heat + humidity + design solutions	2018/2021/2022/2023/2024 envelope family	Core component; time lag/heat capacity are named but not fully derived.
3	Ground/intermediate floor: heat + humidity + DPC/insulation	Same envelope family	Direct component answer with practical damp-proofing response.
4	External/internal walls: orientation, heat, rain/humidity	Same envelope family	High-value wall mechanism + material/thickness/colour/texture response.
5	Window, ventilator, door	Same envelope family	Direct solar gain, infiltration and hot-air exhaust concepts.
6	Balcony, terrace, overhang, clerestory	Envelope family + seasonal-shading short questions partial	Useful sketch-rich support; clerestory is a key ventilation/daylight element.
7	Design tools for moderating microclimate	2018 Q3(c); supports 2022 Q5(a)	Exact list, easy 5-mark answer.
8	Closing slide	No direct question	No study time needed.

Priority note

Slides 2-7 are all exam-relevant. Prepare the answer as a component table: component -> climatic problem -> mechanism -> design solution. This structure fits the repeated 10-22 mark question very well.

2. Detailed Year-wise Question Mapping

Original paper references are preserved. Match labels mean: Exact = the slide block substantially answers the task; Direct = the requested concept is clearly taught but some expansion/sketch/application may still be needed; Partial = the deck only supports part of the requested answer.

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2018	Q3(a)	22	4	Explain how climatic factors affect building-envelope components while designing microclimate; use necessary sketches.	Exact	2-6	This is the central purpose of Lecture 08 and the deck covers each major component with design responses.
2018	Q3(c)	5	4	Mention the design tools for moderating microclimate.	Exact	7	Slide 7 gives the six groups directly.
2021	Q8(a)	12	7	Explain how indoor climatic comfort can be ensured by designing shading devices.	Partial	5-6	The deck covers overhangs, windows and solar-control roles, but shading types/effectiveness are developed in Lecture 09.
2021	Q8(b)	18	7	Explain how climatic factors affect building envelopes; use sketches.	Exact	2-6	Direct repeat of the deck's component-by-component content.
2022	Q5(a)	10	8	Explain designer responsibilities and design tools for moderating microclimate.	Direct	7	The design-tools half is exact; designer-responsibility content belongs to Lecture 04.
2022	Q8(a)	10	9	Explain how climatic comfort can be ensured by shading devices.	Partial	5-6	Window/overhang response is present, but the full shading-device classification belongs to Lecture 09.

2. Detailed Year-wise Question Mapping - Continued

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2022	Q8(b)	14	9	Explain the effects of climatic factors on the building envelope.	Exact	2-6	Direct one-to-one repeat of Lecture 08.
2023	Q6(a)	6	11	Show how an extended overhead/lofty porch blocks summer glare while allowing winter sun.	Partial	6	Overhang solar control is illustrated, but the deck does not explicitly teach the summer-high / winter-low ray construction as a question answer.
2023	Q8(d)	12	12	Explain building-envelope response under climatic factors.	Exact	2-6	Direct repeated envelope family.
2024	Q4(b)	13	14	Explain climatic factors affecting building design and how they influence it.	Direct	2-7	The deck shows how heat, humidity, airflow, solar/rain exposure translate into envelope and design-tool decisions, though the full climatic-data list is wider.
2024	Q5(c)	5	14	Show how an extended lofty porch/overhead shade can stop summer glare but admit winter sun.	Partial	6	Conceptual overhang support only; use the dedicated shading lecture for the seasonal-ray explanation.
2024	Q8(b)	10	15	Explain building-envelope response in microclimate design.	Exact	2-6	Direct repeat of the lecture content.

3. Repeated Question Patterns and Quick Recognition

These patterns compress historical wording into the question families that matter for revision.

P	Normalized repeated question	Years seen	Slides	Why it matters
S	Climatic effects on building-envelope components + design solutions	2018, 2021, 2022, 2023, 2024	2-6	Five-paper repeat and usually 10-22 marks: the dominant reason to master this lecture.
A	Design tools for moderating microclimate	2018; 2022 support	7	Short, direct and easy to score.
A	Window/overhang/clerestory solar and airflow control	2021-2024 related	5-6	Useful overlap with shading and ventilation questions.
B	Time lag / heat-conserving capacity references	2018 related thermal-property questions	2, 4	Named in the deck but not fully derived; do not overstate coverage.

Quick recognition rules

If the question says...	Go to slides	Recognition rule
"building envelope / climatic factors / microclimate"	2-6	Use component-by-component table; do not answer only with general principles.
"roof"	2	Solar heat -> conduction/time lag; humidity -> waterproofing; response -> reflectance, insulation, cavity, roof ventilation.
"wall"	4	Orientation + solar/rain exposure -> material, thickness, colour, texture, openings and rain protection.
"window / ventilator / door"	5	Window = solar/infiltration; ventilator = hot-air exhaust; door = infiltration/access.
"design tools"	7	Materials, orientation, openings, architectural design, micro-planning, semi-mechanized techniques.

4. Slide-to-Question Reverse Index

Use this page when studying directly from the lecture deck: open the slide block, then practice the linked question family immediately.

Slides	What the slides teach	Past-question links	Study action
2	Roof heat/humidity mechanisms + solutions	2018 Q3(a); 2021 Q8(b); 2022 Q8(b); 2023 Q8(d); 2024 Q8(b)	Make one labelled cool/insulated roof sketch.
3	Ground/intermediate floor + DPC/insulation	Same envelope family	Memorize DPC and ground-temperature logic.
4	External/internal walls	Same envelope family	Show orientation/rain arrows + insulation/material response.
5	Window, ventilator, door	Same envelope family	Practice three mini-sketches or a compact table.
6	Balcony, terrace, overhang, clerestory	Envelope family; 2023/2024 seasonal-shading partial	Label sun/rain protection and high-level exhaust.
7	Six design-tool groups	2018 Q3(c); 2022 Q5(a) support	Memorize exact six-item list.
8	Closing	None	Skip.

5. Coverage Gaps and Diagnostic Exclusions

Lecture 08 is very complete for envelope questions, but it does not contain every thermal equation or solar-geometry procedure that appears elsewhere in the question bank.

Gap	Affected questions	Action
Full time-lag, decrement-factor, U-value, thermal-capacity derivations	2018 Q8(b-c) thermal-property family	The deck mentions time lag/heat capacity only as phenomena. Use the dedicated thermal-properties material/reference.
Horizontal/vertical/combined shading classification and effective-shading comparison	2016 Q7(c), Q8(a); 2021/2022 Q8(a)	Use Lecture 09.
Seasonal solar-angle construction and numerical projection sizing	2023 Q6(a), Q7(a); 2024 Q5(c), Q7(b)	Lecture 08 shows an overhang concept but not the full seasonal/numerical method.
Detailed ventilation-rate numericals / ACH / fresh-air table	2021-2024 numerical set	Use Lecture 11.

Mapping discipline	The envelope questions are marked Exact because the deck itself provides component mechanisms and design solutions. By contrast, any calculation or derivation not printed in the deck is kept outside the mapping and listed as a gap.
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6. Recommended Study Plan

This lecture should be studied by envelope component, not slide number alone. Build one master table and attach a small sketch to each major component.

Step	Study block	Slides	Practice immediately
1	Roof + external wall	2, 4	2018 Q3(a); 2021 Q8(b); 2022 Q8(b); 2023 Q8(d); 2024 Q8(b)
2	Floor + window/ventilator/door	3, 5	Same repeated envelope family
3	Balcony/terrace/overhang/clerestory	6	Add to the same master answer; then compare with shading lecture
4	Design tools	7	2018 Q3(c); 2022 Q5(a) support
5	Gap handoff to shading/numericals	6 + Lecture 09/11	2023-2024 shading and ventilation questions

Exam-ready minimum from this lecture

Priority	What to reproduce	Exam technique
A	Envelope master table	Rows: roof, floor, wall, window, ventilator, door, balcony/terrace, clerestory.
A	Roof sketch	Show solar radiation, reflective surface/insulation/cavity and heat-flow arrows.
A	External wall/window sketch	Show solar/rain exposure and protective response.
B	Six design tools	Exact list + one short example each.

Bottom line

Lecture 08 is a must-prepare lecture because one question family alone repeats across five papers. A clean component table plus 2-3 simple labelled sketches can convert these eight slides into a strong 10-22 mark answer.